

## Education

New York University, New York, NY

Aug 2024 – May 2026

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: Machine Learning, Deep Learning, Big Data, System Design, Data Structures

## Experience

- New York University (Novel AI Technologies, Inc.)

Machine Learning Engineer

Aug 2024 – Present

New York, US

– Productionized **computer vision** and **time-series ML models** (**YOLOv8**, **EfficientNet**, **LSTM**, **Isolation Forest**) on cloud inference endpoints; partnered with product and analytics stakeholders to iterate from experiment to deployed behavior.

– Built **Python** and **SQL** ETL pipelines transforming **NoSQL** into **PostgreSQL**; processed **250K+** records and powered **AWS QuickSight** dashboards supporting data-driven operational decisions.

– Developed backend **REST APIs** (**Node.js**, **Python**) for an analytics platform deployed with **Docker** on **EC2**; integrated **LLM-based document analysis** (**LangChain**, **GPT-4**) into underwriting workflows.

• Amazon

Software Development Engineer Intern

May 2025 – Aug 2025

Seattle, US

– Productionized a **RAG pipeline** with an **agentic workflow** and **Model Context Protocol (MCP) server** for operational log retrieval; reduced Athena query time and compute costs by **40%** through optimized inference tooling.

– Deployed event-driven services on **AWS ECS Fargate**, **SQS**, **SNS** with **AWS CDK**; maintained platform reliability with monitoring, testing, and phased rollouts across environments.

• CampusMesh

AI Software Engineer

Jan 2026 – Present

New York, US

– Built and maintained **LLM-based RAG pipelines** with **embedding-based semantic search** for CampusIQ; improved contextual accuracy by **30%** and supported platform evolution across **4 AWS environments**.

– Shipped **Python/Node.js** microservices and backend **REST APIs** on **Lambda**, **API Gateway**, **S3**; containerized releases with **Docker** and **Kubernetes**-style deployment patterns.

• PricewaterhouseCoopers (PwC)

Senior Software Engineer

Aug 2021 – Aug 2024

Bengaluru, India

– Deployed an **NLP-powered OCR neural model** on **Azure** for document extraction in data-intensive healthcare workflows; reduced processing time by **75%** through parallel batch ingestion and model-driven automation.

## Projects

• Flight Delay Prediction (Distributed ML Pipeline) | *PySpark, Spark SQL, HDFS, Python, scikit-learn, SQL*

– Built a scalable **PySpark ML pipeline** on **HDFS** over **7M+** flight records using **Spark SQL** feature engineering; trained **Random Forest** and **Gradient Boosted Tree** models achieving **85%+** accuracy for delay prediction.

• Hospital Management System (Model Serving Platform) | *Python, PyTorch, Hugging Face, AWS SageMaker, MongoDB, REST APIs*

– Fine-tuned a **Hugging Face transformer** model reaching **88% accuracy**; deployed real-time inference on **AWS SageMaker** with backend **REST APIs** and **MongoDB** persistence.

• RoBERTa Fine-Tuning with Custom LoRA | *Python, PyTorch, TensorFlow, Hugging Face, MLflow*

– Architected an end-to-end **ML lifecycle** pipeline with custom **LoRA** fine-tuning, experiment tracking, and evaluation; achieved **83.45%** accuracy with under **1M trainable parameters**.

## Skills

• Programming Languages: Python, SQL, Java, JavaScript, TypeScript

• Machine Learning & Data Science: Machine Learning, Data Science, PyTorch, TensorFlow, scikit-learn, Hugging Face, NLP, Deep Learning, MLflow

• Data & APIs: PySpark, Apache Spark, Spark SQL, REST APIs, Node.js, Pandas, NumPy

• Databases: PostgreSQL, MySQL, MongoDB, DynamoDB

• Cloud & DevOps: AWS (SageMaker, Lambda, ECS, S3, CDK, QuickSight), Docker, Kubernetes, Kafka, CI/CD, Git

## Certifications

AWS Cloud Practitioner

Stanford Machine Learning (Coursera)

Azure Data Fundamentals